Curriculum vitae

Rohit Kumar

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EDUCATION

BS-MS Dual Degree in Biological Sciences **Dec 2020 – June 2025**

Indian Institute of Science Education and Research (IISER) Bhopal, India

Higher Secondary Education (C.B.S.E.) **2017 – 2019**

Varanasi, India

TECHNICAL SKILLS

- **Molecular Biology:** Molecular Cloning (c-DNA and subcloning), Competent cell prep, Plasmid prep, Affinity Protein Purification, RNA and Genomic DNA Isolation, qPCR.
- **Cell Biology & Genetics:** *Drosophila* Genetics and Husbandry (GAL4/UAS, RNAi), Fly Tissue Dissection, Immunofluorescence, Embryo Fixation, Polytene Squash.
- **Biochemistry:** SDS-PAGE, Immunoblotting, Total Protein Extraction & Quantification (Bradford Assay).
- **Microscopy & Imaging:** Confocal Microscopy, Fluorescence Microscopy, Quantitative Analysis (ImageJ).
- **Behavioural Assays:** Larval Crawling Assay, RING Assay.
- **Computational Bioinformatics:** Statistical Analysis (R, GraphPad PRISM), Sequence Analysis (SnapGene), Bioinformatics Tools (BLAST, Clustal Omega, AlphaFold, PyMOL).
- **Coding:** Python , C/C++ (part of curriculum)
- **Scientific Illustration:** Adobe Photoshop, BioRender, and Procreate.

RESEARCH EXPERIENCE

Master's Thesis **May 2024 – June 2025**

Nucleoporin & SUMO Biology Group, IISER Bhopal

Thesis: Functional Insights into Nup43: A Nup107 Complex Member Influencing Locomotory Behavior and Longevity of *Drosophila melanogaster.*

- Validated Nup43's role in *Drosophila melanogaster*'s neurobehavior, and reproductive fitness.
- Confirmed the role of Nup43 in male fertility by rescuing the sterile phenotype, establishing it as a key factor in reproductive development.
- Demonstrated the role of Nup43 in life span of adult *Drosophila melanogaster*.
- Analysed protein localization in gonads, and brains using confocal microscopy.

Winter Research Intern **Dec 2023 – Feb 2024**

Nucleoporin & SUMO Biology Group, IISER Bhopal

Project: Dissect the role of the Nup107-160 nucleoporin complex in *Drosophila melanogaster* development.

- Quantified differential expression of development-associated genes using semi-quantitative PCR and qPCR, linking genotype to molecular phenotypes.
- Screened and validated recombinant fly lines using PCR-based genotyping and Western Blotting to confirm the successful integration and expression of genetic constructs.

Summer Research Intern **May 2023 – Aug 2023**
Nucleoporin & SUMO Biology Group, IISER Bhopal

Project: Investigated the role of Nucleoporin in larval haematopoiesis.

- Quantified circulating haemocytes in third instar larvae to assess hematopoietic defects.
- Knocked down nucleoporin genes and quantified knockdown efficiency using qPCR.

CONFERENCE PRESENTATIONS

Poster presented at:
***11th Annual Biology Meet, IISER Bhopal*, India** **Jan 2025**
J.Kawadkar, Ashley H.Suraj, **Rohit Kumar**
Title: Nup43, a Y-complex member, contributes to efficient reproductive development in *Drosophila melanogaster*.

MANUSCRIPTS IN PREPARATION

J. Kawadkar, A.H. Suraj, R. Kumar, R.K. Mishra.
Essential role for Nup43 in Drosophila fertility and spermiogenesis through Myosin VI-dependent actin cone assembly dynamics
doi: *https://doi.org/10.1101/2025.09.29.679220*

HONORS & COMPETITIVE EXAMINATIONS

- **CSIR-UGC National Eligibility Test (NET), Life Sciences (June 2025)**
  All India Rank : 57

LEADERSHIP & EXTRACURRICULAR ACTIVITIES

Undergraduate Mentor **2024 – 2025 Academic Year**
Mentored three undergraduate students in foundational lab techniques, experimental designs, *Drosophila* genetics, tissue dissection, and immunofluorescence.
Comix Club Coordinator **Oct 2022 – Dec 2023**
Animal Welfare Society Volunteer **Jul 2022 – Mar 2023**

REFERENCES

- **Dr. Ram Kumar Mishra -** Associate Professor, Biological Sciences, IISER Bhopal (Master's Thesis Supervisor , Course instructor)
  Email: rkmishra@iiserb.ac.in